

Moosa Ahmed

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EDUCATION

Wilfrid Laurier University — Waterloo, ON

Expected Dec 2026

Bachelor of Science in Computer Science, Concentration: Software Engineering & Artificial Intelligence

Relevant Coursework: UI/ UX Design, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Systems, Assembly Language, Computer Networks, Software Engineering, Artificial Intelligence, Cryptography, Formal Languages & Automata Theory, Information Retrieval

EXPERIENCE

Backup Supervisor & Storefront Associate

Nov 2024 – Present

Tim Hortons

- Delivered fast, accurate customer service in a high-volume environment, ensuring a positive guest experience during peak hours
- Supervised team operations as Backup Supervisor, delegating tasks and ensuring staff adherence to company standards
- Maintained food safety and cleanliness standards while handling cash transactions and resolving customer concerns

Peer Educator

Jan 2024 – May 2024

Wilfrid Laurier University

- Facilitated academic support sessions for international students, helping them adapt to Canadian post-secondary academic expectations
- Mentored peers on campus resources, study strategies, and navigating college life effectively
- Collaborated with faculty to develop and deliver educational materials, fostering an inclusive learning environment

PROJECTS

Car Rental System | C#, PostgreSQL, Entity Framework

- Built a full-stack web application enabling customers to browse vehicles, make reservations, and manage bookings through a dynamic interface with a secure admin dashboard
- Implemented back-end logic with C# and Entity Framework to handle vehicle availability tracking, pricing calculation, and role-based access control

Parking Assignment: Optimization vs. Reinforcement Learning | Python, Q-Learning, Dijkstra's Algorithm

- Modeled the urban parking assignment problem as both a graph search task and a Markov Decision Process; implemented Dijkstra's algorithm with hill climbing and a Q-learning agent to compare deterministic optimization against adaptive reinforcement learning
- Designed and ran simulations across 3 constraint scenarios (availability, walking-distance limits), evaluating both methods on total cost, adaptability, and convergence; co-authored an 8-page IEEE-format research paper with a 4-person team and presented findings to the class

Digits of e — Digital Logic Circuit | Logic Gates, Circuit Simulation

- Designed a 4-bit combinational logic circuit mapping binary inputs to digits of the mathematical constant e, using truth tables, Karnaugh maps, and minimized Boolean expressions
- Simulated and validated circuit behavior across all 16 input combinations, ensuring correct outputs and proper handling of invalid states

Navigo — Travel & Logistics Platform UX Strategy | UX Design, Figma

- Designed the UX strategy for an all-in-one travel platform concept combining ride-hailing, car rentals, public transit, and food/courier delivery, defining target audience, information architecture, and a 12-page site map
- Conducted competitor research on Uber and Lyft across pricing, UX, and market positioning, and produced a deliverables plan spanning wireframes, mockups, and an interactive prototype

TECHNICAL SKILLS

Languages: Python, Java, C, C#, JavaScript, MATLAB, R, SQL, HTML/CSS, ARM

Tools & Platforms: Git, VS Code, Eclipse, Jupyter, MySQL, PostgreSQL, Linux, CPULator, Figma

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Reinforcement Learning, Database Design, Agile/Scrum

CERTIFICATIONS

- Mastercard: Cybersecurity — Forage (Apr 2026)
- Python Certificate — Kaggle